SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct

I A. Shanbagam / 192426016O (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S Shanbagam . A .

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

CO:2 (Assessment Tool 6)

Application Based Question.

1) Auto - Scaling and Load Balancing.

A company experiences heavy traffic during Seasonal Sales. Auto - Scaling automatically increases the number of Servers when traffic rises and decreases them when traffic is Low. Load balancing distributes user requests evenly across all available Servers preventing any Single Server from becoming overload.

This reduces downtime because if one Server fails, the load balancer redirects traffic to healthy Servers It also improves user experience by providing faster response times.

monitoring tools continuously track CPU usage memory, network traffic, and application performance. When usage crosses a threshold, they trigger auto Scaling.

Example:-

During an Amazon or flipkart Big Billion memory, network traffic, and application adds more Servers to handle millions of users, while load balancing distributes requests to keep the website responsive.

2) Cloud Storage for Backup.

Cloud Storage provides high durability by storing multiple copies of data across different data centers. Data redundancy ensures that if one server fails, another copy is available.

Sensitive backup data is protected using encryption (both during storage and transmission) and access control, where only authorized users can access the data.

Compared to traditional storage, Cloud Storage reduces hardware costs, maintenance costs, and allows users to pay only for the storage they use.

Example :-

A hospital stores patient records in cloud storage. If a flood damages the local servers, the data can be restored from cloud backups, ensuring business continuity.

84 Content Delivery Networks (CDNs)

A Content Delivery Network (CDN) stores copies of website content on edge servers located around the world, users receive content from the nearest server, reducing latency and improving loading speed.

Edge servers reduce the distance data travels improving performance for users in different regions.

Cloud providers support global availability by maintaining data centers across multiple countries ensuring reliable service even during regional failures.

Example :-

Netflix and youtube use CDNs to deliver videos. Smoothly worldwide. Online games like PUBG and free fire edge servers to reduce lag and improve gameplay.

Cloud Security and Compliance.

Cloud providers support regulatory compliance through certifications, security policies, and regular audits.

Logging records all user activities, monitoring detects suspicious behaviour, and encryption protects sensitive financial data from unauthorized access.

Data sovereignty requirements are met by storing customer data in approved geographic regions according to government regulations

Example:-

A bank stores customer transaction data in an Indian cloud region, encrypts all financial records, and maintains audit logs to meet RBI and regulatory compliance requirements.

5) Containerization and Kubernetes

Containers package an application and its dependencies together, making them lightweight and faster than virtual machines because they share the host operating system

Kubernetes automates deployment, scaling, load blancing, and management of containers, making applications highly available.

Containers are portable and can run consistently across AWS, Azure, Google Cloud, or on-premise environments.

Example:-

An e-commerce application uses microservices. where separate containers handle user login, product catalog, payment processing and order management Kubernetes automatically scales these services based on user demand.